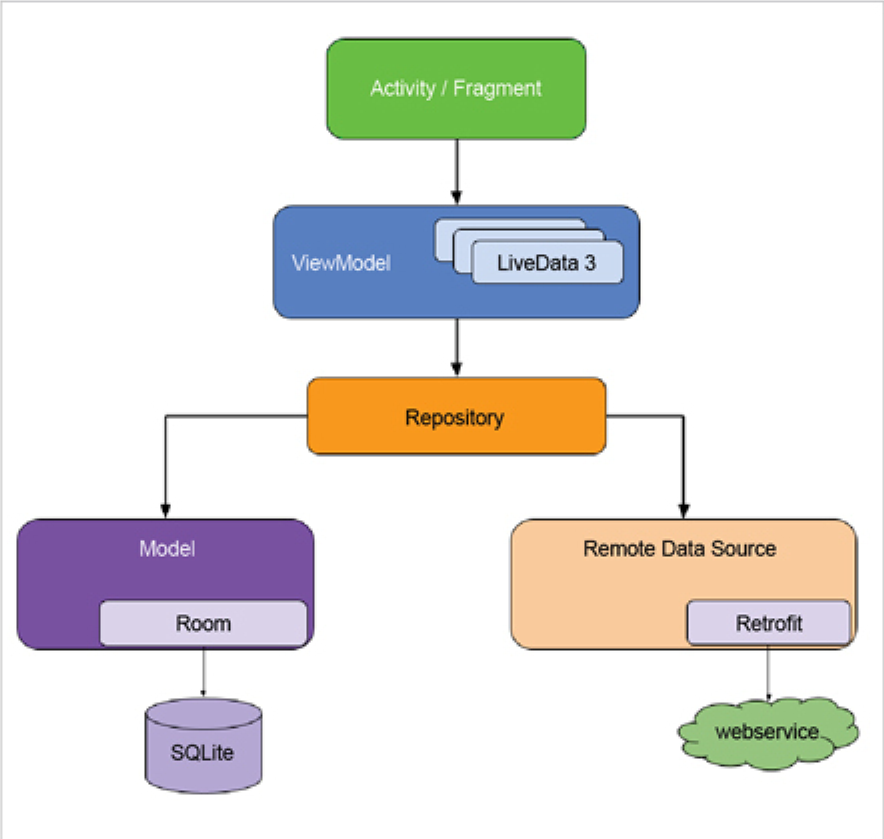

Figure 6: Complete architecture

1. These components make your app modular and focus on ensuring a separation of concerns.
2. They will help you to manage an app's life cycle and tackle those pesky app crashes on rotating the phone.
3. They will help you persist data for building offline apps.
4. They will help you prevent memory leaks and other common issues.
5. And they will prevent you from writing boring boilerplate code.

Final architecture

Google has announced some guidelines with Architecture Components to help build modern Android apps and even recommended which architecture to use for the purpose.

Figure 6 gives the architecture diagram recommended by the Android team in its Architecture Guidelines.

- It suggests keeping Activity and Fragments lean by only maintaining UI related code like click listeners, etc.
- The ViewModel provides the data required by the UI controllers like Activity and Fragments; this helps in surviving configuration change.
- While ViewModel gets the data from the repository, which acts as a single source of truth for the data, it means that whenever your app requires data, it always comes from the repository.
- It's the repository that decides if the data is to be fetched from the local database using Room or from a Web service using Retrofit.

Google recommends the use of Retrofit for networking related code and the use of dependency injection (probably Dagger) to glue all the components.

Table 1 sums up all these features, in a nutshell.

Table 1: The Architecture Guidelines

Component	Action
UI controllers (Activity and Fragment)	Contains only UI related logic
ViewModel	Contains data required by the UI
Repository	Single source of truth for data
Room	Local database
Retrofit	Web service/networking

You can check out my GitHub repository for concrete code and complete apps built using Architecture Components and Guidelines at h*ttps://github.com/AkshayChordiya/android-arch-news-sample/*.

Testing

The Architecture Guidelines help to keep all the components isolated, which is excellent for testing. It's a good practice to test no matter what app you are building. Table 2 gives a ready reckoner to show how to do testing with Architecture Guidelines.

Table 2: How to test with Architecture Guidelines

Component	Test	Mock
UI	Espresso	ViewModel
ViewModel	JUnit	Repository
Repository	JUnit	Room DAO and Retrofit Web Service
Room	Instrumented	-
Retrofit	Instrumented	MockWebServer

Note: Instrumented test cases refer to the test cases that need to run on an Android device.

The Android Architecture Components are amazing independent sets of libraries built to help make Android development easier. With Architecture Guidelines, we get a guide from the Android team on how modern and production-qualityAndroid apps should be built.

Note: There is no perfect architecture for every use case and the architecture mentioned in Architecture Guidelines is just a recommendation. Hence, feel free to use whatever suits your use case the best way.

Libraries like WorkManager and Navigation have recently been added as part of Architecture Components. And more libraries will undoubtedly follow to make Android development easy and fun. END

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